



Inventory Simulation for a Pop-Up Store

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SECTION -- 4B



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Executive Summary

This project mimics inventory operations for a short term pop up retail store running for 7 days. The store retails a small number of rapidly moving consumer products, and experiences problems including limited back-end capacity, a single chance to restock, and heightened variation of consumer demand. Our first aim was to construct a simulation model that could help to make optimized inventory decisions in order for total cost to be minimized and a service level target of over 80% being met. Through construction of a simulation model, we plan to examine how inventory levels react to changing customer demand, determine the risks of overstocking and assess various ways of restocking. The simulation shall help establish an optimal first-inventory and restock policy that optimizes the service levels and holding costs. We as business analytics students intend to apply the concepts of predictive and prescriptive analysis in order to assess risks as regards to the inventory management of the clothing store and prescriptions for possible bottlenecks that might cause below efficient operations to this business. We hope to achieve maximum sales as well as reduce costs by having well defined objectives as demonstrated below.

With Any Logic's discrete-event simulation tools, we simulated this store's daily sales, depletion of inventory and restocking processes. The model included variability in demand, lead times for restocking and physical storage limits. We carried out several sets of parameter variations to determine the most cost effective set of initial inventory, reorder point and restock quantity. After several iterations and analyses, the optimal parameters were found to be:

- **Initial Stock:** 220 units
- **Reorder Point:** 40 units
- **Restock Quantity:** 400 units

This design attained a service level of 84.6% and total cost of \$2822, with a significant improvement, compared to the baseline simulation, which only had a service level of 45.45% and a cost of \$4560.

Problem & Key Performance Indicators (KPIs)

Problem Statement:

- Pop-up store with 7-day window for operations has to improve the inventory management model in order to minimize the level of stockouts and overstocking risk. Customer demand changes daily, and the store's only option in restocking is making a single order with constant lead time of 2 days. The holding and loss sale costs apply to the inventory. The aim is to determine the best inventory policy in which service levels remain high and at their lowest cost to the company.
- Constraints:
 - Inventory capacity limited to 400 units
 - Restock can only be brought once in 7 days.
 - Restock has a fixed 2 day delivery lead time
 - Demand for this simulation is deterministic and known ahead of time

KPIs:

KPI	Description
Service Level (%)	Percentage of customer demand successfully fulfilled
Total Holding Cost (\$)	Sum of daily holding costs (\$1.50 per unit held per day)
Lost Sales Cost (\$)	Sum of lost revenue due to stockouts (\$20 per unfulfilled order)
Total Cost (\$)	Combined holding and lost sales costs

Model Structure

Simulation Tool Used:

- **AnyLogic Personal Learning Edition (PLE)**

Modeling Approach:

A discrete-event model approach was used to construct the simulation model. We turned to AnyLogic's Process Modeling Library in order to build a custom workflow in which customer demand is daily gathered and compared in relation to available inventory. If we go by the inventory levels, restocking sparks off after once during the simulation period.

Key Entities and Parameters:

- **Entities:** RestockOrder agents represent both daily demand and restock events.
- **State Variables:** inventory, dailyDemand, fulfilledSales, lostSales, holdingCost, lostSalesCost, serviceLevel, restocksTriggered, restockOrdered
- **Input Parameters:**
 - initialStock = 220
 - reorderPoint = 40
 - restockQuantity = 400
 - leadTime = 2
 - holdingCostPerUnit = 1.5
 - lostSalesCostPerUnit = 20
 - simulationDays = 7

Process Flow:

1. At the start of each day the demand is generated for a deterministic input.
2. Fulfilled and lost sales are determined subject to availability of inventory.
3. When inventory levels fall below the reorder point, and no restock has been activated, a restock is placed.
4. Restock comes 2 days late and rebuilds the stocks.

5. Co & E – KPIs are calculated for the end of each day - holding cost, lost sales cost, service level.

Results and Graphs

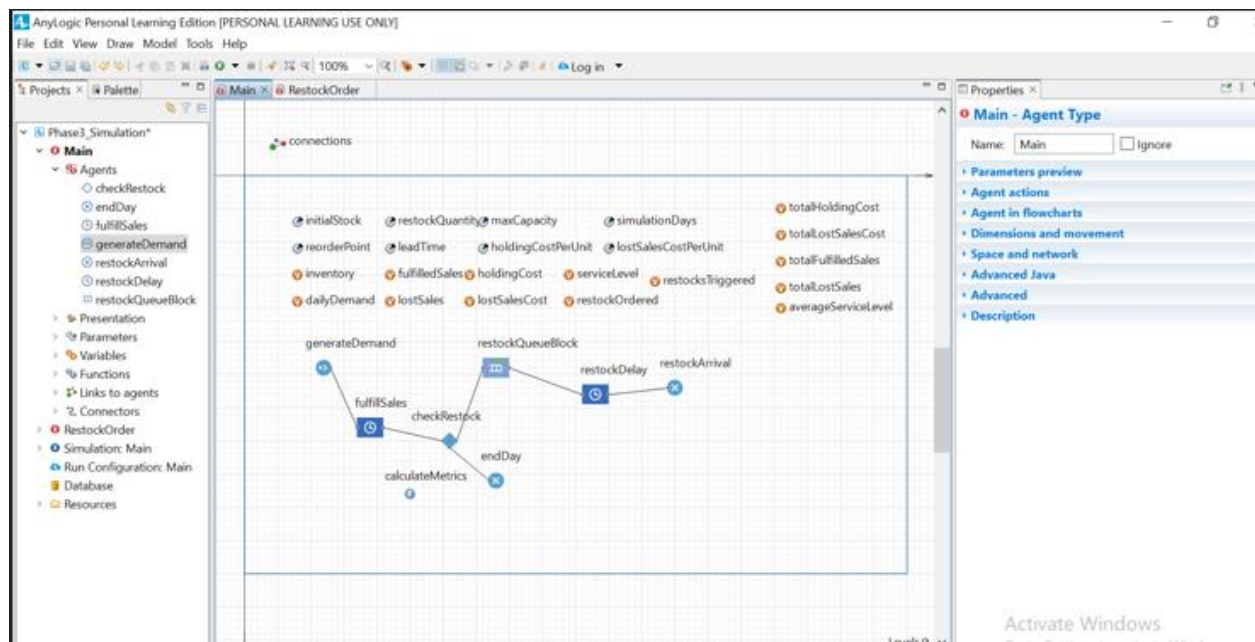
Optimized Simulation Results (Final Run):

Day	Demand	Stock Before Sales	Fulfilled Sales	Lost Sales	Inventory End	Holding Cost (\$)	Lost Sales Cost (\$)	Service Level (%)	Restock Triggered
1	46	220	46	0	174	261	0	100	false
2	59	174	59	0	115	172.5	0	100	false
3	68	115	68	0	47	70.5	0	100	true
4	54	47	47	7	0	0	140	87.04	false
5	54	0	0	54	0	0	1080	0	false
6	47	0	0	0	400	600	0	100	false
7	68	400	68	0	332	498	0	100	false

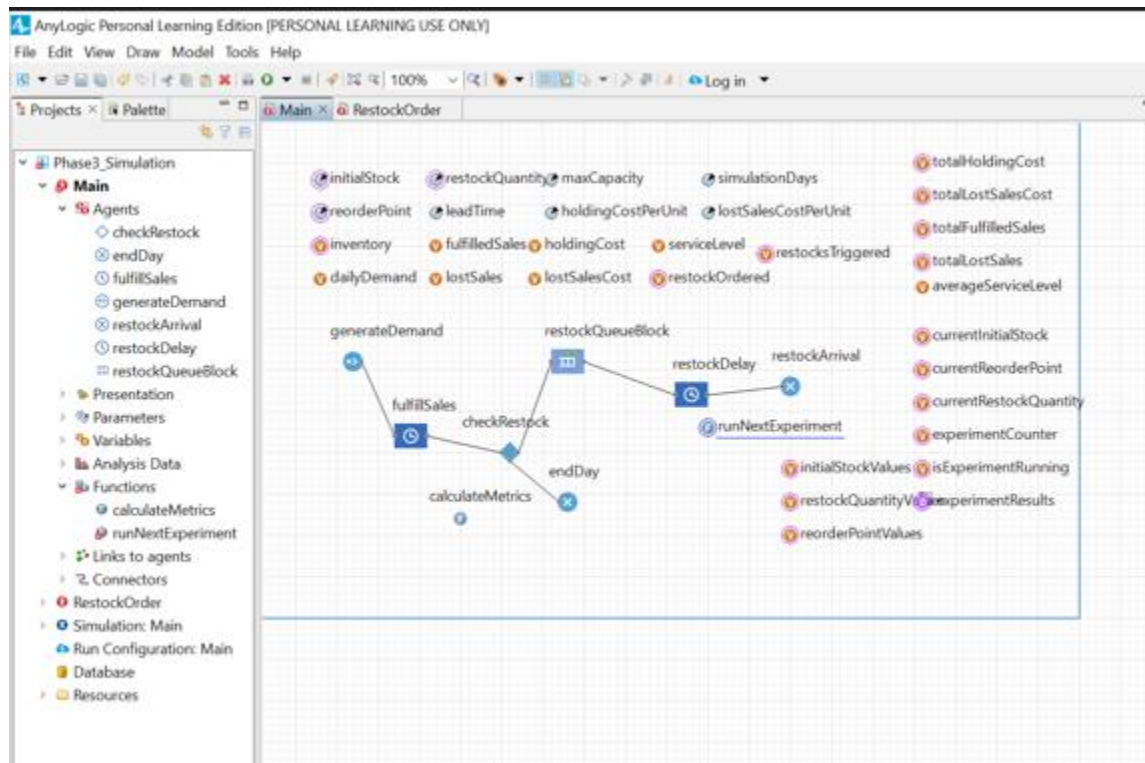
Key Metrics Summary:

- **Total Holding Cost:** \$1602
- **Total Lost Sales Cost:** \$1220
- **Total Cost:** \$2822
- **Average Service Level:** 84.6%

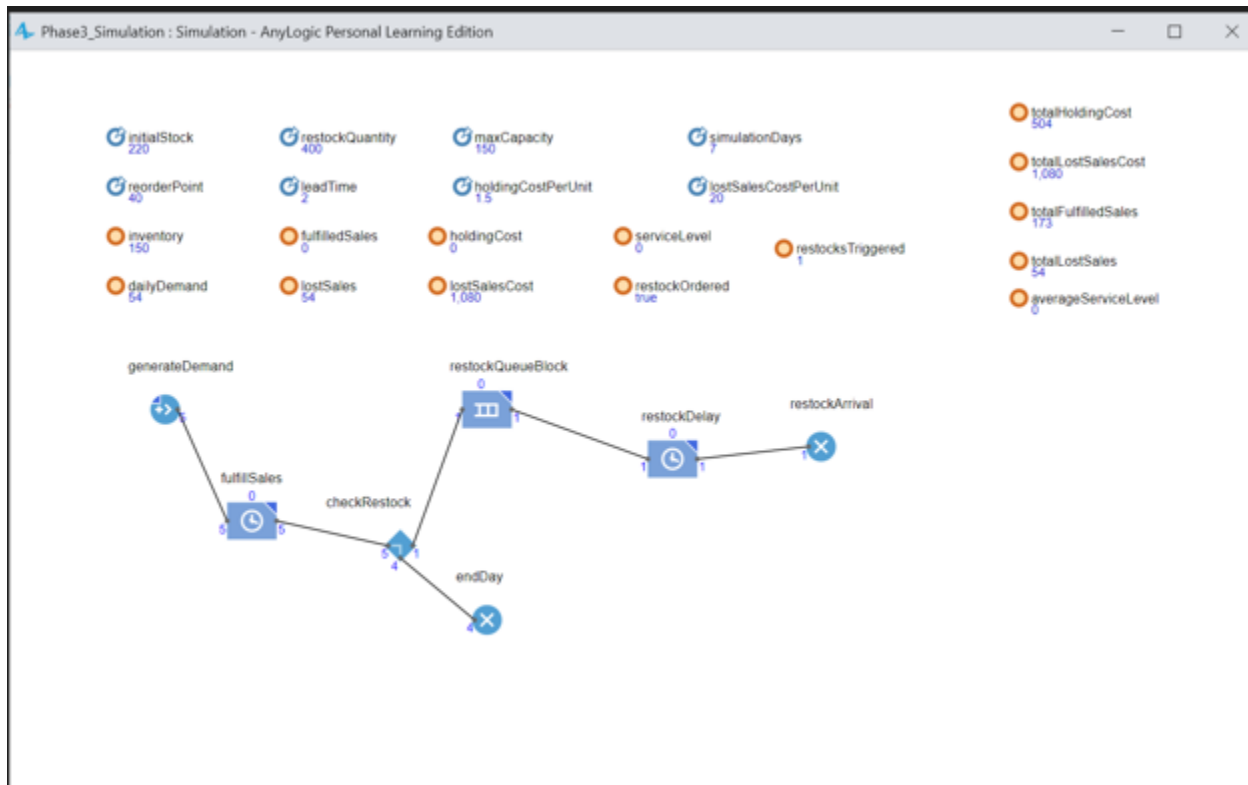
Phase 3 Graph:



Phase 4 Graph:



Phase 5 Graph:



Insights, Recommendations & Ethics

Insights:

- Stocking late can lead to irrecoverable huge lost sales even with high restock quantities.
- Augmentation of initial inventory that leads to additional holding costs can limit other effects of late restocking.
- Using increased reorder point (e.g. 40 units), replenishment becomes early enough to cater for demands spikes.
- Simulation of scenarios before picking inventory levels supports taking data-based choices.

Recommendations:

- Initial Stock – 220, Restock Quantity – 400 to make sure demand is covered up after the initial depletion.
- In future simulations, the introduction of stochastic demand models can be considered to mimic the uncertainty of the real world.
- Give return or donation policies in order to dispose of extra inventory (ethical use)

Ethical Considerations:

- The overstocking can cause waste of resources as well as their unnecessary consumption. Profit, either social or financial, should be compromised with environmental and ethical issues.
- Service level has direct influence on customer satisfaction. Long terms loyalty and brand value should not be bought at the expense of short terms cost cutting.
- Simulations need to be able to apply assumptions without prejudicing their decisions.

Conclusion

This project is able to demonstrate the viability of, not only business analytics but also, business simulation in the solving of real operational issues. Through the Anylogic model, we were able to do inventories strategy simulation, as well as outcomes analysis in conditions of deterministic demand. Through data driven insights we were able to locate an inventory arrangement that is relatively cost effective and meets customer service requirements. The proposed framework can be further extended into other temporary retail settings or modified for probabilistic demand as future work.

The final recommended strategy provides:

- Robust service level of 84.6%
- Cost efficient operations and total cost minimized down to \$2822.
- A scalable model adopted for future experimentation.

We demonstrated using this simulation how inventory optimization can greatly improve business results even in short-term high-variance cases, such as pop-up retail.